where $a_{uu'}^m$ and $b_{uu'}^m$ are the coefficient of $\kappa_m(t)$, and $a_{uu'}^m \in [0,1]$, $b_{uu'}^m \in [-1,0]$. Using multiple basis functions improves the accuracy of model estimation but increases computing complexity. Thus, a commonly used basic function [53] is employed to balance accuracy and complexity of computation. To estimate and handle the parameters in our analysis model, we minimize this following objective function:

$$\min \quad -L(\mu,a,b)+\alpha(\|A\|_F^2+\|B\|_F^2)+\beta(\|A\|_1+\|B\|_1) \tag{5}$$

where $L(\mu,a,b)$ is the Log-Likelihood function, which can be written as follows:

$$L(\mu,a,b)=\sum_{c=1}^{C}\left\{\sum_{i=1}^{N_c}\log\hat{\lambda}_{u_i^c}(t_i^c)-\sum_{u=1}^{U}\int_0^{T_c}\hat{\lambda}_u(s)ds\right\} \tag{6}$$

where μ and a are parameters that must be computed. α and β are hyperparameters that control the regularization terms. The objective function is a convex function [57]. Thus, we choose the Gradient Descent method to minimize Equ 5. Due to the complexity of integral computation, we employ the Monte Carlo method to compute the integral as follows (see more details in supplementary materials):

$$\int_0^{T_c}\hat{\lambda}_u(s)ds=\frac{T_c}{N}\sum_{i=1}^{N}\hat{\lambda}_u\left(t^{(i)}\right) \qquad t^{(i)}\sim U(0,T_c) \tag{7}$$

After obtaining parameters μ, a, and b, the causal strength of entities u on u'can be directly inferred from the impact coefficient a and b. Voting and trading-off strategies are utilized to determine causal strenth by counting the number and the average of a and b in the coefficient. If entity u has an impelling impact on entity u', we statistic the average of positive coefficients in basis function from entity u to entity u'. For example, M_i represent coefficient sets whose basis function coefficients are positive, negative, and zero. M_{max} denotes the set with the largest number of elements among M_i. The causal strength can be inferred as follows:

$$e_{uv}=\frac{1}{|M_{max}|}\sum_{m\in M_{max}}(a_{uv}^m+b_{uv}^m) \qquad M_{max}=\underset{M_i}{\arg\max}\,|M_i| \tag{8}$$

Therefore, we obtain a directed and one-to-one-relationship causality graph $G(V,\ E)$, which unveils impelling and inhibiting behaviors on causality. Edges are weighted by the causal strength e_{uv}, where positive and negative value of e_{uv} indicate impelling and inhibiting behaviors, respectively. Elements in nodes set V are the entities, the edge set E denote the causality between two entities. Then we leverage above causal discovery algorithm, which reveals impelling or inhibiting behavior of causality, to determine the causality of the following candidate cause combinations on effect entity.

4.2 Cause Combination Discovery

A cause combination consists of two or more entities, which individually might not be a cause. The elements in a cause combination interact with each other and jointly have a causal relation with other entity. The causality of these combinations are desired to be uncovered.

However, computing all possible combined causes is infeasible because the number of cause combination sets is exponential. We define "$a \to b$" a individual causality, where a and b refer to a individual cause and effect entity, respectively. A combined causality "$\{a,\ c\} \to b$" is defined, where $\{a,\ c\}$ refers to a cause combination and b is a effect entity. Given the limited entities, the number of possible combined causes is 2^n, where n is the number of all entities. To reduce the number of cause combination, the invalid cause combinations are decreased and useful cause combinations are recruited based on a series of rules.

To guarantee the effectiveness and controllability of candidate cause combinations (T2, T3), the following principles are defined. First, Determining the maximum number of cause combinations is an efficient way and should be evaluated case by case to decrease the number

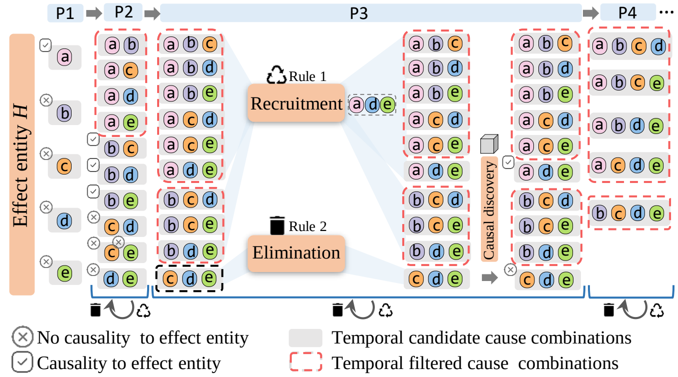

Fig. 3. Illustration of cause combinations process by eliminating the candidate cause combinations and recruiting the filtered combinations with the similarity and co-occurrence of entities.

of candidate cause combinations. The maximum number of cause combinations determine the number of iterations. Second, to further decrease the number of cause combination, cause combination of $\{a,\ c\}$ is regarded as a redundant combination if c have been detected as a cause entity. For example, Figure 3 illustrates cause combination process for a certain effect entity H. If a in P1 is a cause entity of b, we do not consider any cause combination entries that containing a in next iteration. If $\{b,\ c\}$ in P2 is a cause combination of H, we do not consider any cause combination entries containing $\{b,\ c\}$. In this way, the number of candidate cause combination is decreased.

To further improve the effectiveness and usability of candidate cause combinations, we eliminated the invalid cause combinations and recruited the filtered cause combinations. The similarities of entities are acquired by computing the vectors, which are obtained by training temporal event sequence data with Word2Vec [30]. The co-occurrence number of entities are calculated from event sequence dataset. Then, we leverage similarity and chronology co-occurrence of entity for identifying the possibility of cause combination to determine to eliminate or recruit cause combinations. Take procedure P3 of Figure 3 as an example, elimination or recruitment rules are defined.

Eliminating temporal candidate combinations. Candidate cause combinations are obtained shown in the black dotted box after procedure P2, where certain cause combinations may be redundant. Thus, these candidate cause combinations must be further filtered (T3) to decrease computation complexity and improve the effectiveness of cause combination. Considering that similar entities may together facilitate the occurrence of effect entity, we utilize Word2Vec to filter the combinations of irrelevant causes. Given that entities with a low co-occurrence are less likely to be a combination, we calculate the co-occurrence number of entities to filter cause combinations, in which elements have a low co-occurrence.

Recruiting temporal filtered combinations. Redundant cause combinations are filtered shown in the red dotted box after procedure P2, where certain filtered cause combinations may be useful. Therefore, these filtered cause combinations must be further confirmed (T3). Considering that the similar entities may together facilitate the occurrence of the effect entity, we utilize Word2Vec and co-occurrence to detect cause combinations of similar causes. Given that entities with a high co-occurrence tend to be a cause combination, the filtered cause combinations are recruited when the entities in a cause combination have high similarity and high number of co-occurrence.

Algorithm 1 summarizes the process of candidate cause combinations by eliminating the temporal candidate cause combinations and recruiting the temporal filtered cause combinations. The co-occurrence of entities in a candidate cause combination is regarded as a occurrence of a tied entity and fed into our causal discovery algorithm to detect causality. The impelling or inhibiting impact and causal strength of tied entity to effect entity are determined with causal discovery algorithm described in Section 4.1.

After above principles on candidate cause combination and causal discovery, we obtain a combined causal hypergraph $DHG(V,\ DHE)$,